

AI-Based Stepwise Algebra Evaluation Report

Benchmark Evaluation Results: Teacher Ground Truth vs AI System

1. Question-Level Metrics

Metric	Value
MAE	0.8300
Pearson Correlation (r)	0.4810
Exact Match Accuracy	72.00%
Exact Match F1	0.8372
Mean Bias Error (MBE)	-0.7500
RMSE	2.0797
R ² Score	-0.3968

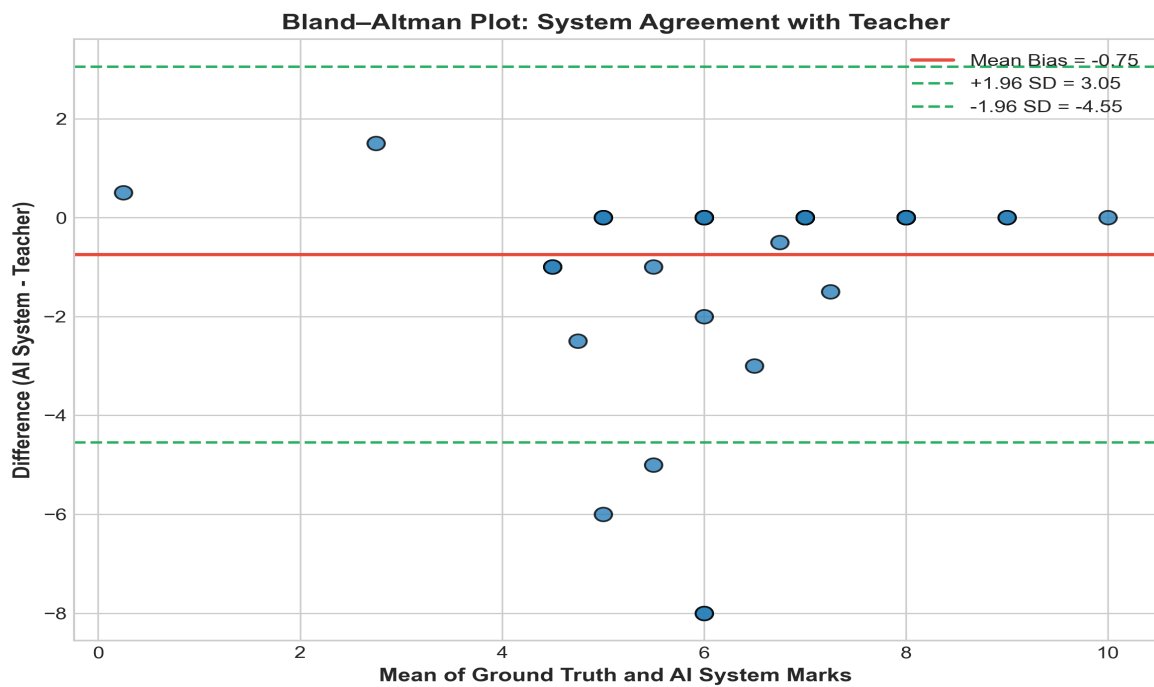
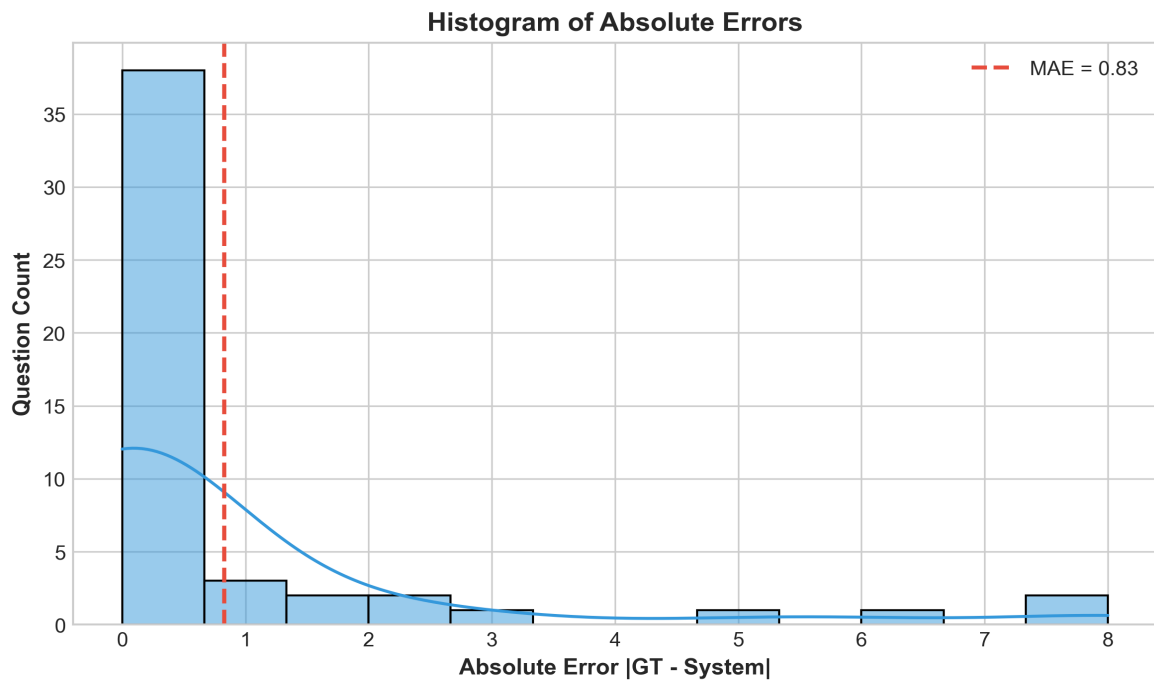
2. Step-Level Evaluation (Validity Classification)

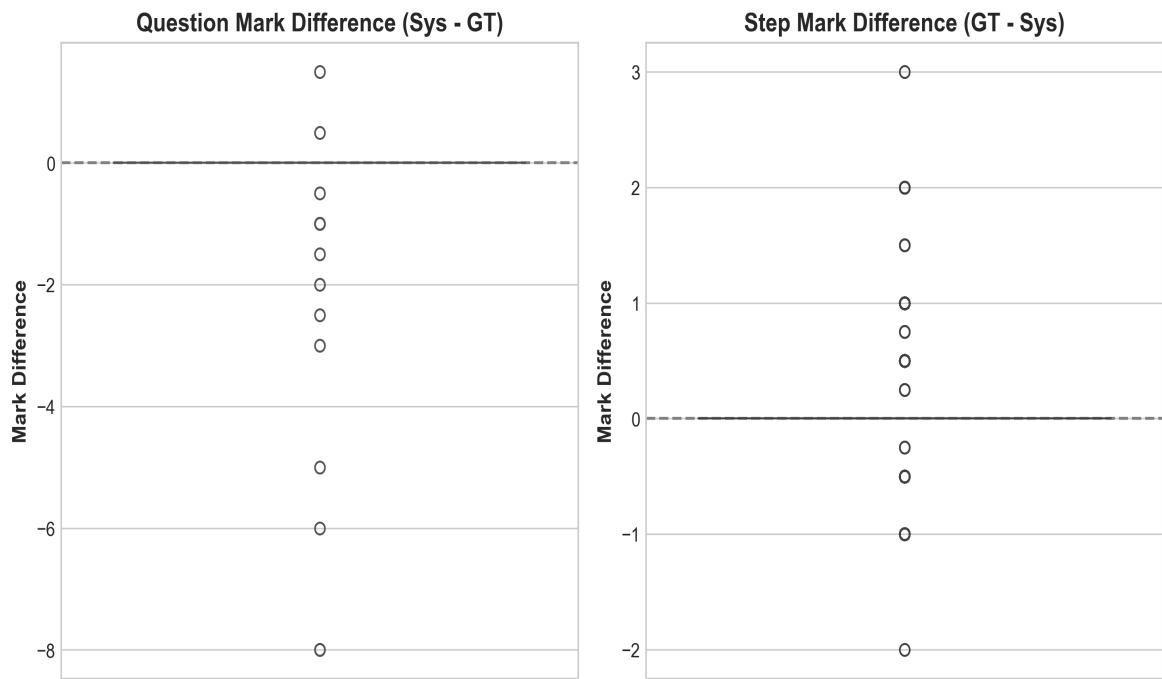
Metric	Value
Macro F1	0.4032
Weighted F1	0.9139
Micro F1	0.8875
F1 (correct)	0.9394
F1 (partially_correct)	0.0000
F1 (incorrect)	0.2703
Cohen's κ	0.1821
Accuracy	88.75%
Precision	0.9496
Recall	0.8875

3. Step Marking Evaluation

Metric	Value
Step MAE	0.3016
Step RMSE	0.5752
Pearson r	0.6082
Mean Difference	-0.1172

4. Visualizations





Step Validity Confusion Matrix			
Ground Truth (Teacher)	correct	incorrect	partially_correct
	correct	incorrect	partially_correct
	279	24	6
	3	5	0
partially_correct	3	0	0
	correct	incorrect	partially_correct
	Predicted (AI System)		

